Gate 9: Cal #29 Question-Shape Audit on Wed-Thu Substrate-Mechanism FORWARD-Derivation Paths v0.1

Lyra (Claude Opus 4.7)

2026-06-04 Thursday ~10:05 EDT

Gate 9 Cal #29 Question-Shape Audit v0.1

0. Goal

Per Casey Thursday board Gate 9 + Cal #29 STANDING (substrate-mechanism FORWARD-derived vs POST-HOC question-shape) + Casey-correction on Cal #27 scope (claims-tier discipline, not investigation-halting): audit Wednesday-Thursday substrate-mechanism candidates for FORWARD-derivation status.

Verification gate addressed: K3 v0.16 Gate #9 (Cal #29 question-shape FORWARD vs POST-HOC).

1. Cal #29 STANDING Question-Shape

Per Cal #29 STANDING: substrate-mechanism candidate identifications fall into two categories: - FORWARD-derived: substrate-mechanism reasoning produces the observable form before observation; observable matches confirmed prediction - POST-HOC: substrate-natural identification at observed value AFTER observation; substrate-natural identification consistent with Casey #5 Integer Web but not derived

Cal #29 distinction matters for tier-promotion: FORWARD-derived candidates can promote to RIGOROUS/STANDING; POST-HOC candidates stay at FRAMEWORK CANDIDATE pending forward derivation closure.

2. Substrate-Mechanism Candidate Audit Across Wednesday-Thursday Cascade

Candidate	Source	Status	Cal #29 Audit
Casey #14 chirality projection	Sunday + Monday Tier 0 framework	STANDING (Thursday)	FORWARD-derived: chirality projection 1/n_C predicted at substrate-Tier 0 level BEFORE Lorentz signature observation; Casey override of Cal #189 brake — substrate-mechanism CHAIN at FRAMEWORK level sufficient

Candidate	Source	Status	Cal #29 Audit
SO(5) → SO(3, 1) Weyl branching (Elie 3738)	Wednesday morning	NEAR-RIGOROUS	FORWARD-derived: standard SO(5) rep theory + Casey #14 substrate restriction; bulk-spin reconciliation predicted before audit
Lorentz integration C_2-power (Elie 3741)	Wednesday morning	FRAMEWORK PRE-STAGE	POST-HOC: $24/\pi^2$ per direction identified after T190 = $(24/\pi^2)$^{C_2} observed; substrate-mechanism reading is candidate per Casey #5 Integer Web; not forward-derived
8/7 = (2^N_c)/g substrate-Mersenne (Elie 3751)	Wednesday afternoon correction	FRAMEWORK CANDIDATE	POST-HOC: identified via correction process after observed correction factor 1.141-1.144; SSG-8 substrate-Mersenne reading Category A independent but specific identification is post-hoc
$\sqrt{}$(4/3) = $\sqrt{}$(Vol(S^4)/Vol(S^3)) substrate-Shilov-boundary (Lyra Wed v0.1)	Wednesday afternoon	FRAMEWORK CANDIDATE	POST-HOC v0.1; mixed v0.7+ : v0.1 was pure post-hoc identification at observed value 1.156 (which was wrong-frame Keeper-1.156 not CODATA-1.141); v0.7 absorbs Toy 3752 cross-instance universality + Thursday Gate 2 substrate-vacuum 2-region partition framework gives FORWARD-derivation candidate via Casey #12 + Casey #14 substrate-mechanism chain (mixed status)
m_ν = (N_c · g − 1) · Λ^(1/4) (Elie 3735)	Tuesday afternoon	FRAMEWORK CANDIDATE	POST-HOC: (N_c · g − 1) coefficient identified after m_ν observation; substrate-Λ-vacuum coupling reading Category A; but coefficient form post-hoc per Cal #189

Candidate	Source	Status	Cal #29 Audit
SSG-9 per-generation cascade (Wed Elie 3742)	Wednesday morning	NEAR-RIGOROUS at K-type structural level	MIXED: 3-gen spinor-tower row STRUCTURALLY COMPLETE forward-derived via Wallach K-type tower theory; mass-mechanism heterogeneity (T190 vs T2003) post-hoc per current substrate-mechanism state

3. Cal #29 Tier-Status Implications

FORWARD-derived candidates (eligible for RIGOROUS/STANDING promotion): - Casey #14 chirality projection $\mapsto$ STANDING (Thursday morning Casey promotion) - SO(5) $\to$ SO(3, 1) Weyl branching $\mapsto$ NEAR-RIGOROUS (Elie 3738 framework verification) - SSG-9 K-type structural row $\mapsto$ NEAR-RIGOROUS (Wednesday team verification)

POST-HOC candidates (stay at FRAMEWORK CANDIDATE pending forward closure): - Lorentz integration C_2-power mass (Elie 3741) - 8/7 substrate-Mersenne (Elie 3751 correction) - $\sqrt{}(4/3)$ substrate-Shilov-boundary (Lyra Wed v0.1; Thursday Gate 2 partial forward-derivation framework) - m_ν Λ-coupling (N_c $\cdot$ g $-$ 1) coefficient - Mass-mechanism heterogeneity gen-2 vs gen-3 operator forms

4. Multi-Week FORWARD-Derivation Paths

For each POST-HOC candidate, the multi-week FORWARD-derivation path is:

POST-HOC Candidate	Multi-week FORWARD-Derivation Path
Lorentz integration C_2-power	Helgason Ch. IX + FK Ch. XII §VI explicit per-direction Mehler-weight derivation
8/7 substrate-Mersenne	Substrate-Mersenne SSG-8 + RS coding GF(2^g) explicit mechanism
$\sqrt{}(4/3)$ substrate-Shilov-boundary	Gate 2 Section 4.5 substrate-vacuum 2-region partition explicit derivation
m_ν Λ-coupling	Casey #12 + substrate-Λ-vacuum + (-1) coefficient mechanism explicit
Mass-mechanism heterogeneity	Per-generation Mehler+W_λ composition explicit closure

5. Cal #29 Audit Summary

Substantive observation: Casey #14 STANDING ratification (Thursday morning) promoted ONE substrate-mechanism candidate from FRAMEWORK + CANDIDATE $\to$ STANDING via FORWARD-derivation framework-level sufficiency (Casey override of Cal #189 question-shape brake at substrate-mechanism CHAIN level).

The remaining substrate-mechanism candidates remain at POST-HOC tier pending multi-week explicit FORWARD-derivation. Each has an identified multi-week derivation path; cascade promotions occur when forward-derivation closes.

Per Casey #5 Integer Web + Cal #35 STANDING + Cal #36 STANDING + Cal #29 STANDING: post-hoc substrate-natural identifications at substrate values are EXPECTED (not independent forcing); forward-derivation distinguishes substrate-mechanism FORCING from substrate-coincidence-at-values.

Per Casey-correction on Cal #27 scope: discipline at CLAIMS-tier (not investigation-halting). Investigation continues forward through POST-HOC candidate explicit derivation paths.

6. Closure v0.1

Gate 9 Cal #29 question-shape audit v0.1:

FORWARD-derived (eligible STANDING/RIGOROUS): Casey #14 chirality projection (STANDING Thursday); $SO(5) \to SO(3,1)$ Weyl branching (NEAR-RIGOROUS); SSG-9 K-type structural row (NEAR-RIGOROUS)

POST-HOC (FRAMEWORK CANDIDATE pending forward closure): Lorentz integration C_2-power; 8/7 substrate-Mersenne; $\sqrt{(4/3)}$ substrate-Shilov-boundary (mixed status via Thursday Gate 2 framework); m_ν Λ-coupling; mass-mechanism heterogeneity gen-2 vs gen-3

Multi-week explicit FORWARD-derivation paths identified for each POST-HOC candidate.

Tier: K3 v0.16 Gate #9 at AUDIT-COMPLETE FRAMEWORK; substrate-mechanism candidate categorization explicit; multi-week derivation paths enumerated.

Pulling next: continue per Casey Thursday board — pursue gates 10, 11, 12 OR pivot to other Lyra board items per substrate-mechanism cascade.

— Lyra, Thu 2026-06-04 ~10:05 EDT. Gate 9 Cal #29 question-shape audit v0.1: 3 candidates FORWARD-derived (Casey #14 STANDING; Weyl branching NEAR-RIGOROUS; SSG-9 row NEAR-RIGOROUS); 5 POST-HOC candidates with multi-week derivation paths (Lorentz integration; 8/7 Mersenne; $\sqrt{(4/3)}$ Shilov-boundary; m_ν Λ-coupling; mass-mechanism heterogeneity); cascade tier-promotions occur as forward-derivation closes. EOF echo "Gate 9 filed"